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Shaft power cycles

The previous chapter has shown that there are many possible arrangements of gas turbines, ranging from the simple cycle, through the regenerative cycle to complex cycles with regeneration, intercooling and reheat. Aircraft engines may appear in the form of turbojets, turboprops, turboshafts or turbofans. The various cycles will be analysed in this and the following chapter. This chapter will deal with land-based shaft power cycles such as those used for electrical power generation or pipeline compression. The next chapter will examine aircraft propulsion cycles, where the variation of forward speed and altitude has a major effect on the performance.

Before proceeding with the main task, the performance prediction of realistic cycles, it is useful to review the performance characteristics of ideal gas turbines in which there are no component inefficiencies or pressure losses. It will be seen that the cycle efficiency depends *only* on the pressure ratio while the specific output depends on *both* the pressure ratio and the turbine inlet temperature. Analysis of ideal gas turbine cycles enables the designer to recognize the importance of particular parameters and provides the foundation for realistic calculations where accurate performance predictions can be made.

2.1 Ideal cycles

Analyses of ideal gas turbine cycles can be found in texts on engineering thermodynamics [e.g. Ref. (1)] and only a brief résumé will be given here. The assumption of ideal conditions will be taken to imply the following:

- (a) Compression and expansion processes are reversible and adiabatic, i.e. isentropic.
- (b) The change of kinetic energy of the working fluid between the inlet and outlet of each component is negligible.
- (c) There are no pressure losses in the inlet ducting, combustion chambers, heat-exchangers, intercoolers, exhaust ducting and ducts connecting the components.

- (d) The working fluid has the same composition throughout the cycle and is a perfect gas with constant specific heats.
- (e) The mass flow of gas is constant throughout the cycle.
- (f) Heat transfer in a heat-exchanger (assumed counterflow) is 'complete', so that in conjunction with (d) and (e) the temperature rise on the cold side is the maximum possible and exactly equal to the temperature drop on the hot side.

Assumptions (d) and (e) imply that the combustion chamber, in which fuel is introduced and burned, is considered as being replaced by a heater with an external heat source. For this reason, as far as the calculation of performance of ideal cycles is concerned, it makes no difference whether one is thinking of them as 'open' or 'closed' cycles. The diagrammatic sketches of plant will be drawn for the normal case of the open cycle.

Simple gas turbine cycle

The ideal cycle for the simple gas turbine is the Joule (or Brayton) cycle, i.e. cycle 1234 in Fig. 2.1. The relevant steady flow energy equation is

$$Q = (h_2 - h_1) + \frac{1}{2}(C_2^2 - C_1^2) + W$$

where Q and W are the heat and work transfers per unit mass flow. Applying this to each component, bearing in mind assumption (b), we have

$$W_{12} = -(h_2 - h_1) = -c_p(T_2 - T_1)$$

$$Q_{23} = (h_3 - h_2) = c_p(T_3 - T_2)$$

$$W_{34} = (h_3 - h_4) = c_p(T_3 - T_4)$$

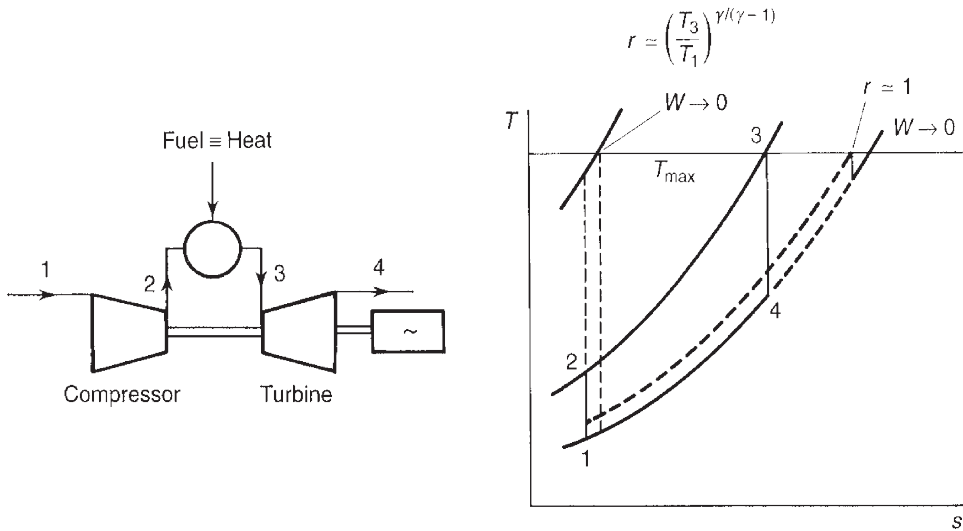


FIG. 2.1 Simple cycle

The cycle efficiency is

$$\eta = \frac{\text{net work output}}{\text{heat supplied}} = \frac{c_p(T_3 - T_4) - c_p(T_2 - T_1)}{c_p(T_3 - T_2)}$$

Making use of the isentropic p - T relation, we have

$$T_2/T_1 = r^{(\gamma-1)/\gamma} = T_3/T_4$$

where r is the pressure ratio $p_2/p_1 = r = p_3/p_4$. The cycle efficiency is then readily shown to be given by

$$\eta = 1 - \left(\frac{1}{r}\right)^{(\gamma-1)/\gamma} \quad (2.1)$$

The efficiency thus depends only on the pressure ratio and the nature of the gas. Figure 2.2(a) shows the relation between η and r when the working fluid is air ($\gamma = 1.4$), or a monatomic gas such as argon ($\gamma = 1.66$). For the remaining curves in this section air will be assumed to be the working fluid. The effect of using helium instead of air in a closed cycle is studied in section 2.6, where it is shown that the theoretical advantage indicated in Fig. 2.2(a) is not realized when component losses are taken into account.

The specific work output W , upon which the size of plant for a given power depends, is found to be a function not only of pressure ratio but also of maximum cycle temperature T_3 . Thus

$$W = c_p(T_3 - T_4) - c_p(T_2 - T_1)$$

which can be expressed as

$$\frac{W}{c_p T_1} = t \left(1 - \frac{1}{r^{(\gamma-1)/\gamma}} \right) - (r^{(\gamma-1)/\gamma} - 1) \quad (2.2)$$

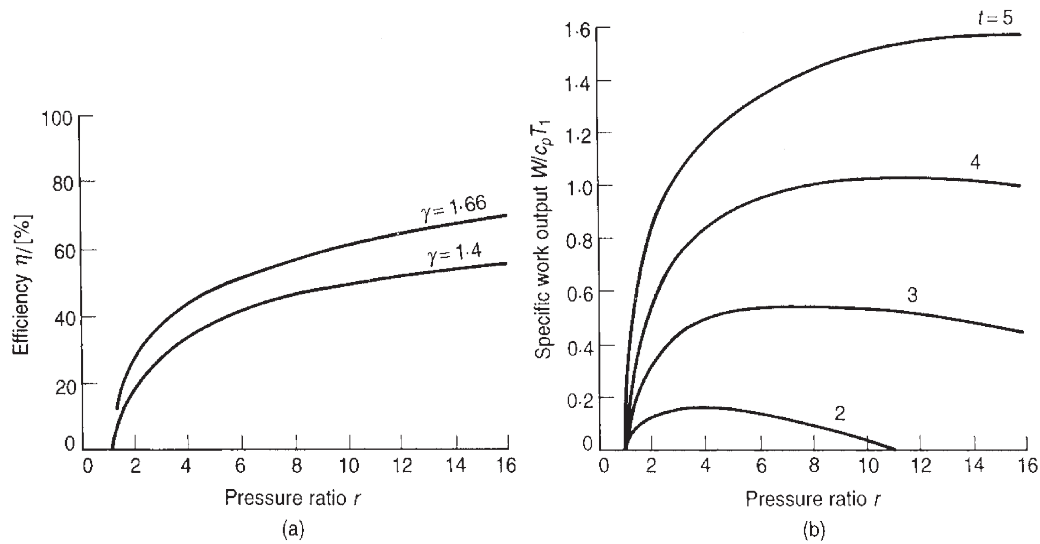


FIG. 2.2 Efficiency and specific work output—simple cycle

where $t = T_3/T_1$; T_1 is normally atmospheric temperature and is not a major independent variable. It is therefore convenient to plot the specific work output in non-dimensional form ($W/c_p T_1$) as a function of r and t as in Fig. 2.2(b). The value of T_3 , and hence t , that can be used in practice depends upon the maximum temperature which the highly stressed parts of the turbine can stand for the required working life: it is often called the 'metallurgical limit'. Early gas turbines used values of t between 3.5 and 4, but the introduction of air-cooled turbine blades allowed t to be raised to between 5 and 6.

A glance at the T - s diagram of Fig. 2.1 will show why a constant t curve exhibits a maximum at a certain pressure ratio: W is zero at $r = 1$ and also at the value of r for which the compression and expansion processes coincide, namely $r = t^{\gamma/(\gamma-1)}$. For any given value of t the optimum pressure ratio for maximum specific work output can be found by differentiating equation (2.2) with respect to $r^{(\gamma-1)/\gamma}$ and equating to zero. The result is

$$r_{\text{opt}}^{(\gamma-1)/\gamma} = \sqrt{t}$$

Since $r^{(\gamma-1)/\gamma} = T_2/T_1 = T_3/T_4$, this is equivalent to writing

$$\frac{T_2}{T_1} \times \frac{T_3}{T_4} = t$$

But $t = T_3/T_1$ and consequently it follows that $T_2 = T_4$. Thus the specific work output is a maximum when the pressure ratio is such that the compressor and turbine outlet temperatures are equal. For all values of r between 1 and $t^{\gamma/2(\gamma-1)}$, T_4 will be greater than T_2 and a heat-exchanger can be incorporated to reduce the heat transfer from the external source and so increase the efficiency.

Heat-exchange cycle

Using the nomenclature of Fig. 2.3, the cycle efficiency now becomes

$$\eta = \frac{c_p(T_3 - T_4) - c_p(T_2 - T_1)}{c_p(T_3 - T_5)}$$

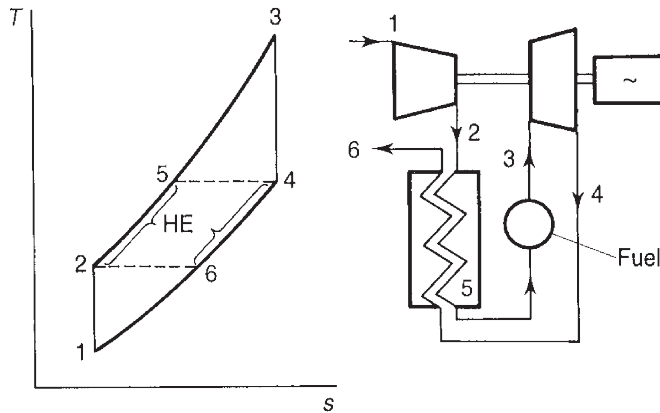


FIG. 2.3 Simple cycle with heat-exchange (HE)

With ideal heat-exchange $T_5 = T_4$, and on substituting the isentropic p - T relations the expression reduces to

$$\eta = 1 - \frac{r^{(\gamma-1)/\gamma}}{t} \quad (2.3)$$

Thus the efficiency of the heat-exchange cycle is not independent of the maximum cycle temperature, and clearly it increases as t is increased. Furthermore it is evident that, for a given value of t , the efficiency increases with decrease in pressure ratio and not with increase in pressure ratio as for the simple cycle. The full lines in Fig. 2.4 represent the equation, each constant t curve starting at $r = 1$ with a value of $\eta = 1 - 1/t$, i.e. the Carnot efficiency. This is to be expected because in this limiting case the Carnot requirement of complete external heat reception and rejection at the upper and lower cycle temperature is satisfied.

The curves fall with increasing pressure ratio until a value corresponding to $r^{(\gamma-1)/\gamma} = \sqrt[t]{t}$ is reached, and at this point equation (2.3) reduces to (2.1). This is the pressure ratio for which the specific work output curves of Fig. 2.2(b) reach a maximum and for which it was shown that $T_4 = T_2$. For higher values of r a heat-exchanger would *cool* the air leaving the compressor and so reduce the efficiency, and therefore the constant t lines have not been extended beyond the point where they meet the efficiency curve for the simple cycle which is shown dashed in Fig. 2.4.

The specific work output is unchanged by the addition of a heat-exchanger and the curves of Fig. 2.2(b) are still applicable. From these curves and those in Fig. 2.4 it can be concluded that to obtain an appreciable improvement in efficiency by heat-exchange, (a) a value of r appreciably less than the optimum for maximum specific work output should be used and (b) it is not necessary to use a higher cycle pressure ratio as the maximum cycle temperature is increased. Later it will be shown that for real cycles conclusion (a) remains true but conclusion (b) requires modification.

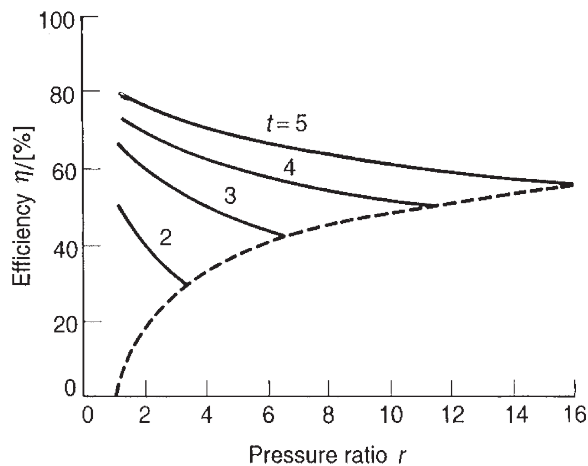


FIG. 2.4 Efficiency—simple cycle with heat-exchange

Reheat cycle

A substantial increase in specific work output can be obtained by splitting the expansion and reheating the gas between the HP and LP turbines. Figure 2.5(a) shows the relevant portion of the reheat cycle on the T – s diagram. That the turbine work is increased is obvious when one remembers that the vertical distance between any pair of constant pressure lines increases as the entropy increases: thus $(T_3 - T_4) + (T_5 - T_6) > (T_3 - T'_4)$.

Assuming that the gas is reheated to a temperature equal to T_3 , differentiation of the expression for specific work output shows that the optimum point in the expansion at which to reheat is when the pressure ratios (and hence temperature drops and work transfers) for the HP and LP turbines are equal. With this optimum division, it is then possible to derive expressions for the specific output and efficiency in terms of r and t as before. Writing $c = r^{(\gamma-1)/\gamma}$ they become

$$\frac{W}{c_p T_1} = 2t - c + 1 - \frac{2t}{\sqrt{c}} \quad (2.4)$$

$$\eta = \frac{2t - c + 1 - 2t/\sqrt{c}}{2t - c - t/\sqrt{c}} \quad (2.5)$$

Comparison of the $W/c_p T_1$ curves of Fig. 2.6 with those of Fig. 2.2(b) shows that reheat markedly increases the specific output. Figure 2.5(b), however, indicates that this is achieved at the expense of efficiency. This is to be expected because one is adding a less efficient cycle (4'456 in Fig. 2.5(a)) to the simple cycle—less efficient because it operates over a smaller temperature range. Note that the reduction in efficiency becomes less severe as the maximum cycle temperature is increased.

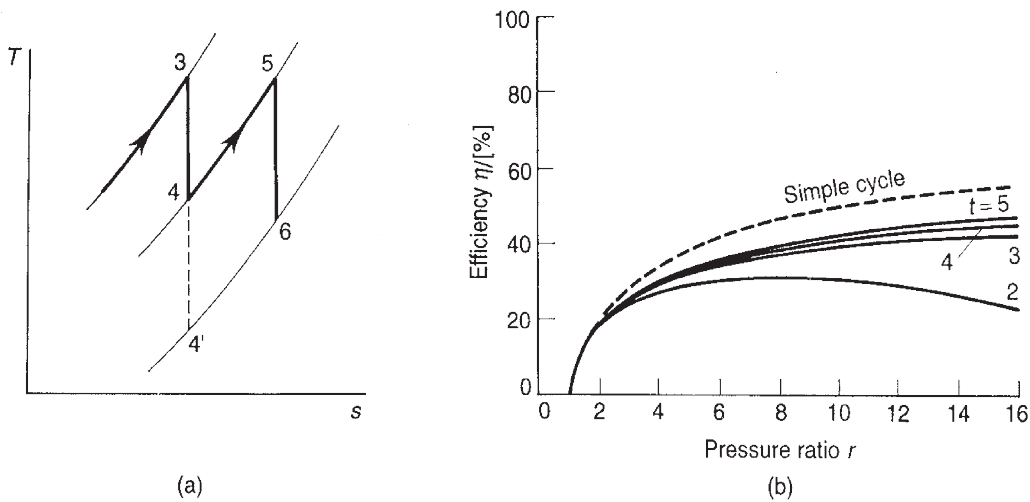


FIG. 2.5 Reheat cycle

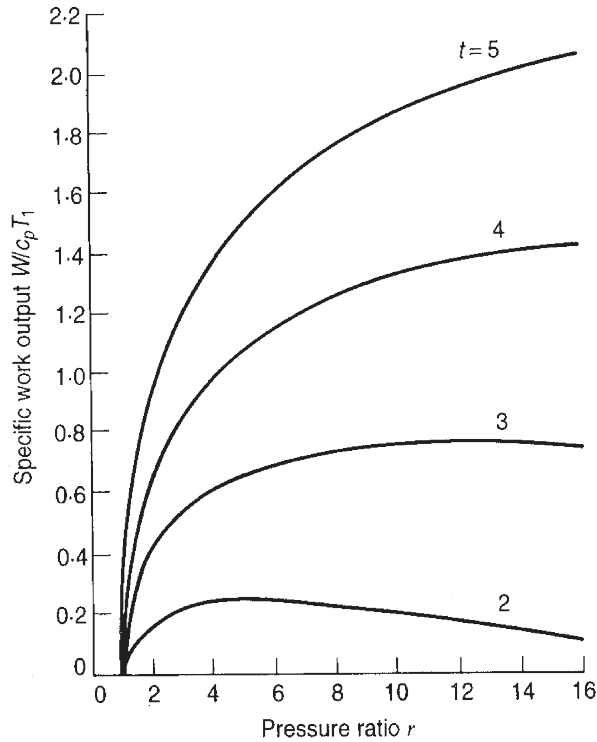


FIG. 2.6 Work output-reheat cycle

Cycle with reheat and heat-exchange

The reduction in efficiency due to reheat can be overcome by adding heat-exchange as in Fig. 2.7. The higher exhaust gas temperature is now fully utilized in the heat-exchanger and the increase in work output is no longer offset by the increase in heat supplied. In fact, when a heat-exchanger is employed, the efficiency is higher with reheat than without as shown by a comparison of Figs 2.8 and 2.4. The family of constant t lines exhibit the same features as those for the simple cycle with heat-exchange—each curve having the Carnot value at $r = 1$ and falling with

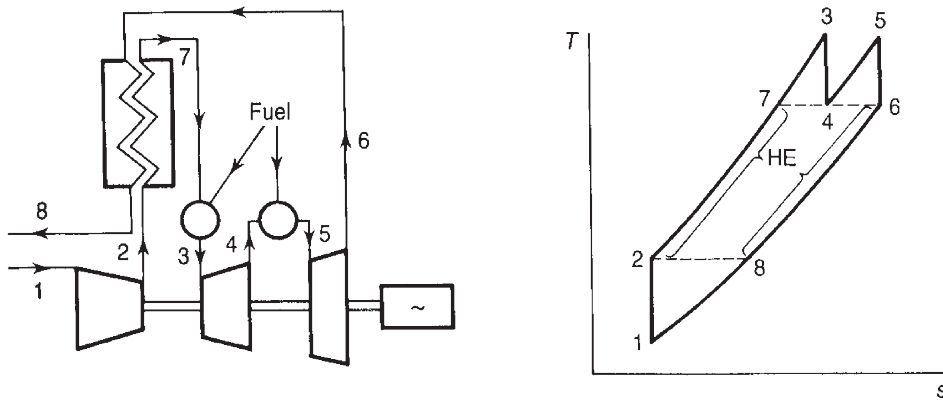


FIG. 2.7 Reheat cycle with heat-exchange

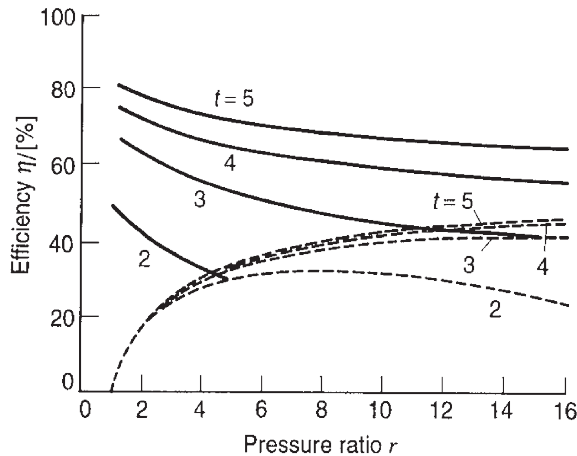


FIG. 2.8 Efficiency—reheat cycle with heat-exchange

increasing r to meet the corresponding efficiency curve of the reheat cycle without heat-exchange at the value of r corresponding to maximum specific work output.

Cycles with intercooled compression

A similar improvement in specific work output to that obtained by reheat can be achieved by splitting the compression and intercooling the gas between the LP and HP compressors; and, assuming that the air is intercooled to T_1 , it can be shown that the specific work output is a maximum when the pressure ratios of the LP and HP compressors are equal. The use of intercoolers is seldom contemplated in practice because they are bulky and need large quantities of cooling water. The main advantage of the gas turbine, that it is compact and self-contained, is then lost. For this reason no performance curves for cycles with intercooling are included: suffice it to say that they are similar to Figs 2.5(b) and 2.6, although the increase in specific work output and reduction in efficiency with respect to the simple cycle are not so marked. (A modification to the low-temperature region of a cycle is normally less significant than a comparable modification to the high-temperature region.) As with reheat, intercooling increases the cycle efficiency only when a heat-exchanger is also incorporated, and then curves almost identical to those of Fig. 2.8 are obtained.

This discussion of ideal cycles should be sufficient to indicate the main effects of additions to the simple cycles gas turbine. The simple equations developed would never be used in practice, but correctly show that use of a high pressure ratio is the key to high thermal efficiency, while high turbine inlet temperature is essential for high specific output. When dealing with real cycles, in the next section, it will be seen that both parameters are important for high efficiency. The simple theory also shows that regenerative cycles require a lower pressure ratio for high efficiency, but increasing maximum cycle temperature results in an increased pressure ratio for best efficiency.